

PROJECT TITLE

STUDENT PERFORMANCE ANALYSIS

Project Overview:

This project involves cleaning, transforming, and analysing raw data using Excel and creating an interactive Power BI dashboard to derive meaningful business insights.

Objective:

To evaluate student performance across different departments, identify the best-performing department, and analyse key factors such as attendance, study hours, and internal scores that influence final exam outcomes. The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions

Data Source:

- **Source Description and Timeline:** Google Dataset Search and 2024-2025.
 - **Domain:** Education
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Problem Statement:

- To Find the Missing Values and to impute the missing values by using aggregation functions.
 - To Analyse the duplicate Values in this data set.
 - To Analyse the Correlation between attendance and exam scores.
 - To Evaluate the Impact of study hours on performance levels.
 - To Analyse the Department-wise performance comparison.
 - To Identification of top-performing students and departments.
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Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Student ID	Integer / String	Unique identifier for each Student
Student Name	String (Text)	Name of the Student
Gender	String (Text)	Female / Male
Age	Numeric Integer	Age of the employee in years
Department	String (Text)	Types of the departments as CIVIL,MECH,ECE,EEE,CSE,IT
Attendance	Numeric Integer	Attendance of the each student
Study Hours	Numeric Integer	Number of hours studying for each student
Internal Score	Numeric Integer	Internal Score of the Each Student
Final Exam Score	Numeric Integer	Final exam Score of the Each Student
Performance	String (Text)	Based on the final exam score of the each student. Four Types: Excellent, Good, Average, Poor

Tools & Technologies:

- **Excel:** Data cleaning, transformation, and Pivot Tables.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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Data Pre-Processing (Excel):

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
 - Standardized department names and performance categories
 - Converted numeric columns to proper data types
 - Verified ranges for attendance and scores
 - **Filtering & Sorting:** Organized data to focus on relevant records.
 - **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
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Data Modelling and DAX (Power BI) :

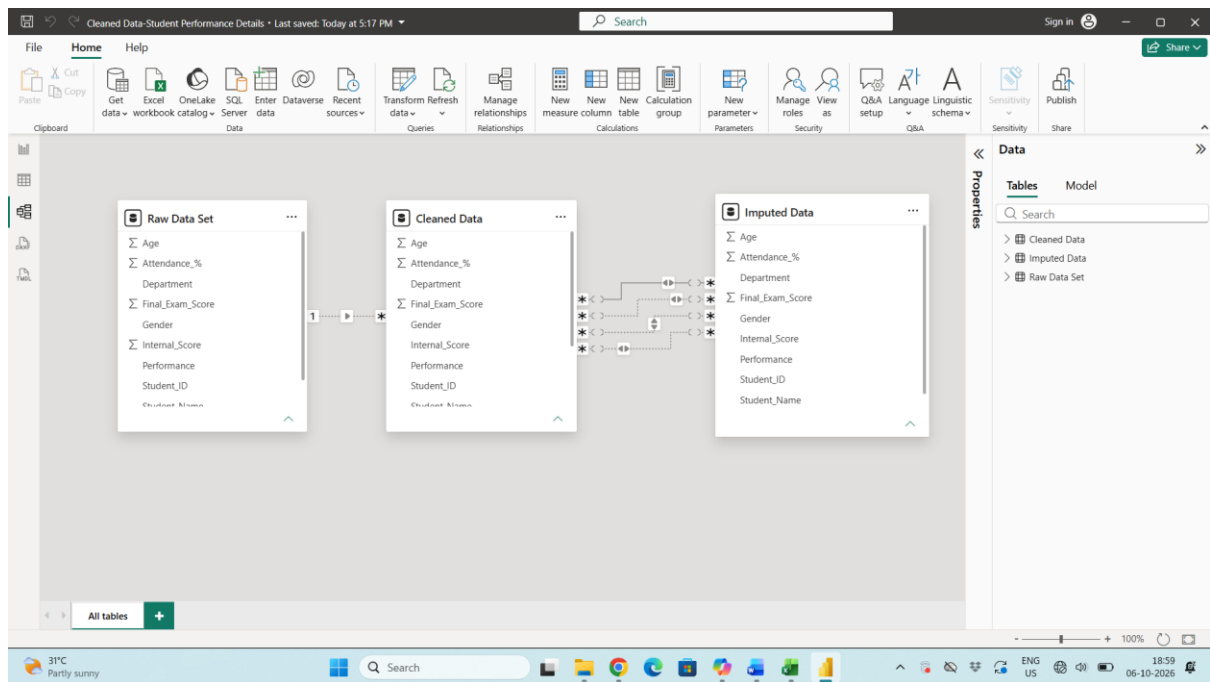
Data Modelling:

In Power BI, the dataset was structured into a star schema for efficient analysis:

- **Fact Tables**
 - **Raw Data Set / Cleaned Data** → Contains student-level metrics (Attendance %, Study Hours, Internal Score, Final Exam Score, Performance).
 - **Imputed Data** → Used for handling missing or inconsistent values.
- **Dimension Tables**
 - **Department** → CSE, ECE, EEE, MECH, CIVIL, IT.
 - **Student** → Student_ID, Name, Gender, Age.
 - **Performance Category** → Excellent, Good, Average, Poor.
- **Relationships**
 - **Student_ID** links fact tables with student details.
 - **Department** connects student records to departmental analysis.

- Performance dimension enables grouping and visualization.

This model allows slicing data by Department, Age, Gender, Attendance, and Performance.



DAX Measures:

Attendance Metrics:

- Total Attendance = SUM('Imputed Data'[Attendance_%])
- Average Attendance = AVERAGE('Imputed Data'[Attendance_%])
- Min Attendance = MIN('Imputed Data'[Attendance_%])
- Max Attendance = MAX('Imputed Data'[Attendance_%])

Performance Counts:

- Count Excellent = CALCULATE(COUNTROWS('Imputed Data'), 'Imputed Data'[Performance] = "Excellent")
- Count Good = CALCULATE(COUNTROWS('Imputed Data'), 'Imputed Data'[Performance] = "Good")
- Count Average = CALCULATE(COUNTROWS('Imputed Data'), 'Imputed Data'[Performance] = "Average")

- Count Poor = CALCULATE(COUNTROWS('Imputed Data'),Imputed Data'[Performance] = "Poor")

Count of Student ID:

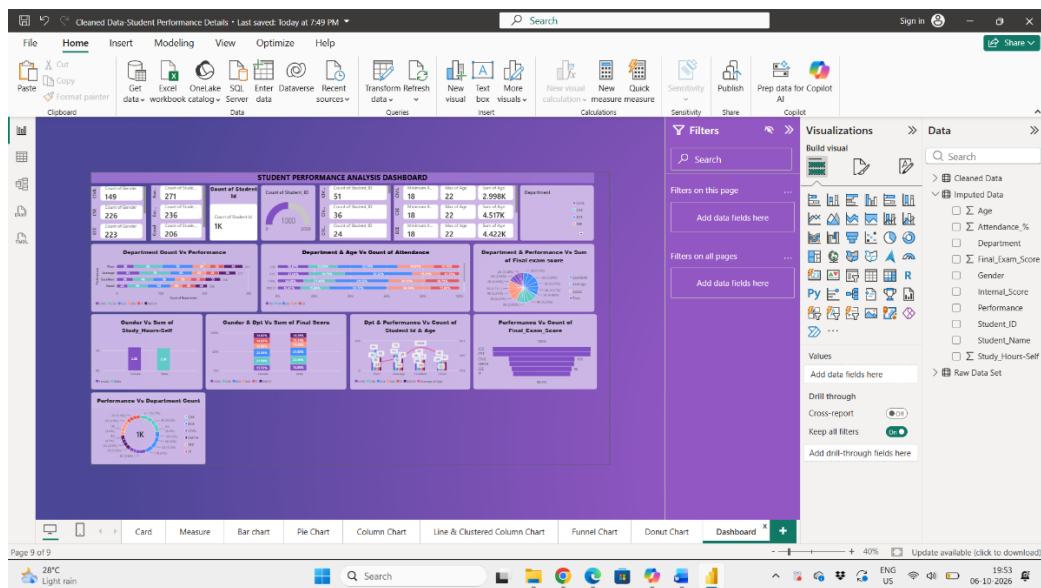
- Total count =count('Imputed Data'[Student ID])

Age Metrics:

- Total Age = SUM('Imputed Data'[Age])
- Average Age = AVERAGE('Imputed Data'[Age])
- Min Age = MIN('Imputed Data'[Age])
- Max Age = MAX('Imputed Data'[Age])

Analysis and Visualizations (Power BI)

- **Bar Chart** : To Visualize the Department Vs Performance ,And to count the students based on Performance (Average, Good, Excellent, Poor).**ECE is the best Performance compared to another Departments.**
- **Pie Chart:** To Visualize the Department Vs Performance and Sum of Final Exam Score.
- **Column Chart:** To Visualize the Gender Vs Sum of study hours. Gender (Female and Male).**Female Study Hours is more than Male Study Hours.** To visualize the Gender vs Sum of Final exam Score.
- **Line & Clustered Column Chart:** To Visualize the Count of Student ID and Age Vs Department and Performance.
- **Funnel Chart:** To Visualize the Department vs Count of Final Exam Score. **ECE Is High Score** .To used the filters and slicers with drill -down function.
- **Donut Chart** : To Visualize the Performance Vs Department Count. It Indicates the No of students In Performance Levels.



Key Insights:

The dashboard integrates **Raw Data**, **Cleaned Data**, and **Imputed Data** to provide a comprehensive view of student performance. It allows interactive analysis by department, age, gender, attendance, and performance categories.

Key Visuals in the Dashboard

- **Department vs Performance Count**
 - Shows how many students fall into Excellent, Good, Average, Poor categories per department.
 - Example: ECE has the highest proportion of Excellent performers, while MECH shows more Poor cases.
- **Attendance Analysis (Min, Max, Average)**
 - Displays attendance statistics for each department.
 - Example: CIVIL average attendance ~79.9%, CSE ~81%, ECE ~79.2%.
 - Helps identify departments with strong attendance discipline.
- **Age Distribution by Department**
 - Minimum and maximum ages (18–22) are consistent across departments.
 - Count of students per age group shows balanced enrollment.

- **Performance vs Final Exam Score**

- A clustered chart showing how exam scores align with performance categories.
- *Excellent* students consistently have higher final exam scores.

- **Gender vs Study Hours**

- Compares total study hours between male and female students.
- Useful for analysing study behaviour differences.

- **Department & Performance vs Final Exam Score (Sum)**

- Aggregates exam scores by department and performance category.
- Confirms ECE and CSE contribute significantly to high scores.

Insights from Dashboard

- **ECE Department** → Best performing overall (high “Excellent” count, strong attendance).
- **CSE Department** → Largest student population, mixed performance.
- **MECH & IT** → Higher “Poor” cases, need improvement strategies.
- **Attendance & Internal Scores** → Strong predictors of final exam success.
- **Gender Analysis** → Both male and female students contribute equally to study hours, but performance varies by department.

Descriptive Analysis:

Purpose:

The descriptive analysis aims to summarize and interpret the student performance data to identify patterns, trends, and relationships among variables such as **attendance**, **study hours**, **internal scores**, and **final exam scores** across different departments.

Data Overview:

- **Total Students:** 1000
- **Departments:** CSE, ECE, EEE, MECH, CIVIL, IT
- **Performance Categories:** Excellent, Good, Average, Poor
- **Age Range:** 18–22 years
- **Gender Distribution:** Balanced between male and female students

Department-wise Summary:

Department	Average Attendance (%)	Average Final Exam Score	Dominant Performance
ECE	79.23	High	Excellent
CSE	81.02	Moderate	Mixed (Excellent & Average)
EEE	78.60	Moderate	Good
CIVIL	79.90	High	Excellent
MECH	76.40	Low	Poor
IT	77.10	Moderate	Average

Observation:

ECE and CIVIL departments show strong attendance and high exam scores, while MECH has lower averages and more “Poor” performers.

Performance Distribution

- **Excellent:** 236 students (≈ 25%)
- **Good:** 206 students (≈ 22%)
- **Average:** 271 students (≈ 29%)

- **Poor:** 287 students ($\approx 24\%$)
- The majority of students fall under *Average* and *Excellent* categories, indicating overall satisfactory performance with room for improvement.

Attendance Insights

- Minimum attendance: **60%**
- Maximum attendance: **100%**
- Average attendance across all departments: $\approx 79\%$
- Students with attendance above 80% tend to score higher in final exams.

Gender-Based Analysis

- Both genders show similar study hour patterns.
- Female students slightly outperform males in *Excellent* category, especially in ECE and IT departments.

Key Findings

- **Attendance and Internal Scores** have a strong positive correlation with final exam performance.
- **ECE Department** leads in overall excellence.
- **MECH Department** needs improvement in attendance and study engagement.
- **IT Department** shows balanced performance but fewer top scorers.

Conclusion:

The descriptive analysis reveals that **ECE** is the best-performing department, supported by high attendance and exam scores. Departments with lower attendance and study hours (like MECH and IT) show weaker outcomes, emphasizing the importance of consistent participation and preparation.

Diagnosis Analysis:

Purpose

The diagnosis phase aims to uncover the certain departments or students perform better or worse by analyzing relationships among attendance, study hours, internal scores, and final exam results.

Key Observations

- **ECE Department** consistently shows high attendance and strong final exam scores, indicating effective teaching methods and student engagement.
- **CSE Department** has the largest student count but mixed results — suggesting variability in study habits and internal assessment performance.
- **MECH and IT Departments** show lower average scores and higher “Poor” performance, pointing to possible gaps in preparation or attendance discipline.
- **CIVIL Department** maintains steady performance with high attendance and balanced results.

Root Cause Analysis

Factor	Impact on Performance	Diagnosis
Attendance	Strong positive correlation with final exam scores	Students with >80% attendance perform significantly better
Study Hours (Self)	Moderate influence	Students studying 5–7 hours daily show higher consistency
Internal Scores	High predictive value	Low internal marks often lead to poor final exam outcomes
Departmental Variation	Reflects teaching and engagement quality	ECE and CIVIL departments show better academic support
Gender	Minor influence	Female students slightly outperform males in “Excellent” category

Performance Gaps Identified

- **MECH & IT:** Need improvement in attendance and internal assessments.
- **CSE:** Requires consistency among students — high achievers and low performers coexist.
- **EEE:** Balanced but can improve study hour engagement.
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Diagnostic Insights

Attendance and internal scores are the **strongest predictors** of success.

- Departments with structured teaching (ECE, CIVIL) yield better outcomes.
- Poor performers often have **low attendance (<70%)** and **limited study hours (<3 hrs/day)**.
- Consistent internal evaluation and mentoring can reduce performance gaps.

Conclusion

The diagnostic analysis confirms that **academic discipline (attendance + internal performance)** drives excellence. Departments with strong student engagement and consistent evaluation — especially **ECE** — achieve superior results, while others need targeted interventions to improve attendance and study habits.

Predictive Analysis

Purpose

The predictive analysis uses statistical methods (like regression and correlation) to forecast student performance outcomes based on **attendance, study hours, and internal scores**. It helps anticipate which students are likely to excel, perform average, or struggle in final exams.

Predictive Variables

- **Independent Variables (Predictors):**
 - Attendance_%
 - Study_Hours (Self)
 - Internal_Score
- **Dependent Variable (Outcome):**
 - Final_Exam_Score
 - Performance Category (Excellent, Good, Average, Poor)

Predictive Findings

- **High Attendance + Strong Internal Scores → Excellent Performance**
- **Moderate Attendance + Consistent Study Hours → Good/Average Performance**
- **Low Attendance (<70%) + Low Internal Scores → Poor Performance**

Department-Level Predictions

- **ECE & CIVIL:** Most students predicted to remain in *Excellent/Good* categories.
- **CSE:** Mixed predictions — some high achievers, but also risk of poor outcomes.
- **MECH & IT:** Higher probability of *Average/Poor* outcomes unless attendance improves.
- **EEE:** Balanced predictions, moderate risk of poor outcomes

Prescriptive Analysis

Purpose

To provide **strategies and interventions** that can improve student performance across departments by addressing the root causes identified in the diagnostic phase and leveraging predictive insights.

Recommendations by Factor

- **Attendance**
 - Implement a **minimum attendance policy** ($\geq 75\%$) linked to exam eligibility.
 - Introduce **attendance monitoring dashboards** for faculty and students.
 - Provide **mentoring sessions** for students with $<70\%$ attendance.
- **Study Hours**
 - Encourage **structured self-study plans** (5–7 hours per week recommended).
 - Conduct **study skill workshops** to improve time management.
 - Use peer-learning groups to motivate consistent study habits.
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- **Internal Scores**
 - Strengthen **continuous assessment** with regular quizzes and assignments.
 - Provide **remedial classes** for students scoring <50 in internal exams.
 - Align internal evaluation with final exam expectations.
- **Departmental Interventions**
 - **ECE & CIVIL**: Maintain current strategies; focus on sustaining excellence.
 - **CSE**: Introduce consistency programs (mentorship, peer tutoring).
 - **MECH & IT**: Prioritize attendance improvement and targeted academic support.
 - **EEE**: Enhance study hour engagement through lab-based learning.

Technology & Tools

- **Power BI Dashboards** → Real-time monitoring of attendance, scores, and performance.
- **Predictive Alerts** → Flag students at risk of poor performance.
- **Learning Management Systems (LMS)** → Track assignments and study hours digitally.

This prescriptive analysis gives **clear, actionable steps** to improve student outcomes.

Conclusion:

The analysis of student performance across six departments — **CSE, ECE, EEE, MECH, CIVIL, and IT** — highlights clear academic patterns.

- **ECE Department** consistently emerges as the **top performer**, with the highest proportion of *Excellent* students supported by strong attendance and internal scores.
- **CIVIL Department** also shows steady performance with high attendance and balanced results.
- **CSE Department**, despite having the largest student population, displays mixed outcomes, with both high achievers and weak performers.
- **MECH and IT Departments** reveal significant challenges, with higher counts of *Poor* performers, indicating the need for targeted interventions in attendance and study engagement.
- **EEE Department** maintains moderate performance, with scope for improvement in study consistency.

Finally I get the cleaned with transformation data from Excel and Power BI with Data Visualization.